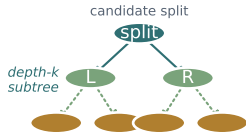


Sparse farther-sighted trees improve forest tradeoffs

k-sighted induction evaluates splits through optimized local descendants.

Bounded non-greedy splits



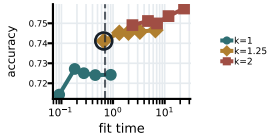
Split choice optimizes impurity over a bounded local subtree.

Greedy defaults stay competitive

- Single trees gain only thin accuracy improvements.
- Uniformly farther-sighted forests rarely repay their time cost.
- A few $k=2$ trees in a greedy forest can lift accuracy at similar fit time.

Extra sight helps little on average; sparse sight inside forests is the exception.

Sparse sight helps forests



Mixed forests outperform uniform ones at equal computation time.

Takeaway: greedy trees are strong; use extra sight sparsely inside forests.